

1 Resit coursework solutions

1. [5 marks] What are the Fourier coefficients c_k of $\sin^4 x$?

$$\begin{aligned}(\sin x)^4 &= \left(\frac{\exp(ix) - \exp(-ix)}{2i} \right)^4 \\&= \left(\frac{\exp(2ix) - 2 + \exp(-2ix)}{-4} \right)^2 \\&= \frac{\exp(4ix) - 4\exp(2ix) + 6 - 4\exp(-2ix) + \exp(-4ix)}{16}\end{aligned}$$

hence, $c_{-4} = c_4 = 1/16$, $c_{-2} = c_2 = -1/4$, $c_0 = 3/8$ and $c_k = 0$ otherwise.

2. [5 marks] Show for $0 \leq k, \ell \leq n-1$

$$\frac{1}{n} \sum_{j=1}^n \cos k\theta_j \cos \ell\theta_j = \begin{cases} 1 & k = \ell = 0 \\ 1/2 & k = \ell \\ 0 & \text{otherwise} \end{cases}$$

for $\theta_j = \pi(j-1/2)/n$. Hint: You may consider replacing $\cos$ with complex exponentials:

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}.$$

We have,

$$\begin{aligned}\frac{1}{n} \sum_{j=1}^n \cos(k\theta_j) \cos(\ell\theta_j) &= \frac{1}{4n} \sum_{j=1}^n \left(e^{i(k+l)\theta_j} + e^{-i(k+l)\theta_j} + e^{i(k-l)\theta_j} + e^{-i(k-l)\theta_j} \right) \\&= \frac{1}{4n} \sum_{j=1}^n \left(e^{ia_{kl}\theta_j} + e^{-ia_{kl}\theta_j} + e^{ib_{kl}\theta_j} + e^{-ib_{kl}\theta_j} \right),\end{aligned}$$

where we have defined $a_{kl} = k+l$ and $b_{kl} = k-l$. Now consider, for $a \in \mathbb{Z}$, $a \neq 2kn$ for some $k \in \mathbb{Z}$,

$$\begin{aligned}\sum_{j=1}^n e^{ia\theta_j} &= \sum_{j=1}^n e^{ia\pi(j-\frac{1}{2})/n} \\&= e^{-ia\pi/2n} \sum_{j=1}^n e^{iaj\pi/n} \\&= e^{-ia\pi/2n} \sum_{j=1}^n (e^{ia\pi/n})^j \\&= e^{-ia\pi/2n} e^{ia\pi/n} \frac{(e^{ia\pi/n})^n - 1}{e^{ia\pi/n} - 1} \\&= e^{ia\pi/2n} \frac{e^{ia\pi} - 1}{e^{ia\pi/n} - 1},\end{aligned}$$

where our assumptions on a ensure that we are not dividing by 0. Then we have, for a as above,

$$\begin{aligned}
\sum_{j=1}^n (e^{ia\theta_j} + e^{-ia\theta_j}) &= e^{ia\pi/2n} \frac{e^{ia\pi} - 1}{e^{ia\pi/n} - 1} + e^{-ia\pi/2n} \frac{e^{-ia\pi} - 1}{e^{-ia\pi/n} - 1} \\
&= e^{ia\pi/2n} \frac{e^{ia\pi} - 1}{e^{ia\pi/n} - 1} + e^{-ia\pi/2n} \cdot \frac{e^{ia\pi/n}}{e^{ia\pi/n}} \cdot \frac{e^{-ia\pi} - 1}{e^{-ia\pi/n} - 1} \\
&= e^{ia\pi/2n} \frac{e^{ia\pi} - 1}{e^{ia\pi/n} - 1} + e^{ia\pi/2n} \frac{e^{-ia\pi} - 1}{1 - e^{ia\pi/n}} \\
&= e^{ia\pi/2n} \frac{e^{ia\pi} - 1}{e^{ia\pi/n} - 1} - e^{ia\pi/2n} \frac{e^{-ia\pi} - 1}{e^{ia\pi/n} - 1} \\
&= \frac{e^{ia\pi/2n}}{e^{ia\pi/n} - 1} (e^{ia\pi} - 1 - e^{-ia\pi} + 1) \\
&= \frac{e^{ia\pi/2n}}{e^{ia\pi/n} - 1} \frac{1}{2i} \sin(a\pi),
\end{aligned}$$

which is 0 for a an integer.

Now, when $k = l = 0$, we have $a_{kl} = b_{kl} = 0$, and,

$$\frac{1}{n} \sum_{j=1}^n \cos(k\theta_j) \cos(l\theta_j) = \frac{1}{4n} \sum_{j=1}^n (1 + 1 + 1 + 1) = 1.$$

When $k = l \neq 0$, we have $0 < a_{kl} = 2k < 2n$, and $b_{kl} = 0$. Hence,

$$\begin{aligned}
\frac{1}{n} \sum_{j=1}^n \cos(k\theta_j) \cos(l\theta_j) &= \frac{1}{4n} \sum_{j=1}^n (e^{ia_{kl}\theta_j} + e^{-ia_{kl}\theta_j} + 1 + 1) \\
&= \frac{1}{4n} \left[\sum_{j=1}^n (e^{ia_{kl}\theta_j} + e^{-ia_{kl}\theta_j}) + 2n \right] \\
&= \frac{1}{2},
\end{aligned}$$

since a_{kl} meets the conditions for the sum considered above.

When $k \neq l$, we have, $-2n < a_{kl}, b_{kl} < 2n$ and $a_{kl}, b_{kl} \neq 0$, hence,

$$\begin{aligned}
\frac{1}{n} \sum_{j=1}^n \cos(k\theta_j) \cos(l\theta_j) &= \frac{1}{4n} \sum_{j=1}^n (e^{ia_{kl}\theta_j} + e^{-ia_{kl}\theta_j} + e^{ib_{kl}\theta_j} + e^{-ib_{kl}\theta_j}) \\
&= \frac{1}{4n} \left[\sum_{j=1}^n (e^{ia_{kl}\theta_j} + e^{-ia_{kl}\theta_j}) + \sum_{j=1}^n (e^{ib_{kl}\theta_j} + e^{-ib_{kl}\theta_j}) \right] \\
&= 0.
\end{aligned}$$

3. **[5 marks]** Consider the Discrete Cosine Transform (DCT)

$$C_n := \begin{bmatrix} \sqrt{1/n} & & & \\ & \sqrt{2/n} & & \\ & & \ddots & \\ & & & \sqrt{2/n} \end{bmatrix} \begin{bmatrix} 1 & \cdots & 1 \\ \cos \theta_1 & \cdots & \cos \theta_n \\ \vdots & \ddots & \vdots \\ \cos(n-1)\theta_1 & \cdots & \cos(n-1)\theta_n \end{bmatrix}$$

for $\theta_j = \pi(j-1/2)/n$. Prove that C_n is orthogonal: $C_n^\top C_n = C_n C_n^\top = I$. Hint: $C_n C_n^\top = I$ might be easier to show than $C_n^\top C_n = I$ using the previous problem.

The components of C without the diagonal matrix, which we may call $\hat{C}$ are

$$\hat{C}_{ij} = \cos((j-1)\theta_{i-1}),$$

where $\theta_j = \pi(j-1/2)/n$. Recalling that the elements of matrix multiplication are given by

$$(ab)_{ij} := \sum_{k=1}^n a_{ik} b_{kj}$$

we find that

$$(\hat{C}_n \hat{C}_n^\top)_{ij} = \sum_{k=1}^n \cos((i-1)\theta_{k-1}) \cos((k-1)\theta_{j-1}).$$

By using the previous problem and the terms on the diagonal matrices which ensure that the $1/2$ terms become 1 we know how to compute all of these entries and find that it is the identity.

Here is a computer-based demonstration:

```
using LinearAlgebra, Plots, FFTW
n = 5
θ = range(π/(2n); step=π/n, length=n) # n evenly spaced points starting at π/(2n)
with step size π/n
C = Diagonal([1/sqrt(n); fill(sqrt(2/n), n-1)]) * [cos((k-1)*θ[j]) for k=1:n, j=1:n]
C'C
```

```
5×5 Matrix{Float64}:
 1.0 -4.85266e-18 -2.82901e-17 1.68455e-17 3.95658e-17
-4.85266e-18 1.0 4.68569e-17 -6.18283e-17 4.0019e-17
-2.82901e-17 4.68569e-17 1.0 6.05122e-18 -1.76076e-16
1.68455e-17 -6.18283e-17 6.05122e-18 1.0 1.03351e-16
3.95658e-17 4.0019e-17 -1.76076e-16 1.03351e-16 1.0
```

4. **[10 marks]** Consider the variable-coefficient advection equation

$$u_t + c(x)u_x = 0, \quad c(x) = \frac{1}{5} + \sin^2(x-1), \quad x \in [0, 2\pi], \quad t \in [0, T],$$

with $u(x, 0) = f(x) = e^{-100(x-1)^2}$, which we approximate with the leapfrog method

$$\mathbf{u}^{i+1} = \mathbf{u}^{i-1} - 2\tau \mathcal{F}^{-1} \left\{ i(-m:m) \cdot \mathcal{F}\{\mathbf{u}^i\} \right\}, \quad i = 0, \dots, n_t - 1.$$

Note that one needs $\mathbf{u}^{-1}$ to initialise the leapfrog method. Let the entries of $\mathbf{u}^{-1}$ be $u_j^{-1} = f(x_j + \tau/5)$, $j = 0, \dots, n_x - 1$. The exact solution is periodic in time, i.e.,

$$u(x, t + T) = u(x, t)$$

where

$$T = \int_0^{2\pi} \frac{1}{c(x)} dx = 12.8254983016186 \dots$$

Compute T using the Trapezoidal rule and confirm that you get the value stated above. Then compute

$$e(n_x) = \max_{j=0, \dots, n_x-1} |u_j^0 - u_j^{n_t}|$$

where $n_t = 8n_x$ and $\tau n_t = T$ and plot $e(n_x)$ for $n_x = 2^k + 1$, with $k = 5, 6, \dots, 10$. Comment on the behaviour of $e(n_x)$.

As discussed in the notes, the trapezoidal rule approximation to an integral is

$$\int_a^b g(x) dx \approx \frac{h}{2} (g(x_0) + 2g(x_1) + 2g(x_2) + \dots + 2g(x_{n-1}) + g(x_n)).$$

To compute T , we set $a = 0$, $b = 2\pi$ and $g(x) = 1/c(x)$ and note that since $c(x)$ is 2π -periodic, $g(x_0) = g(x_n)$, hence

$$\int_0^{2\pi} \frac{1}{c(x)} dx \approx \frac{2\pi}{n} \sum_{j=0}^{n-1} \frac{1}{c(x_j)}$$

```
c = x -> 0.2 + sin(x - 1)^2
n = 80
xx = range(0, 2*pi; length=n+1)[1:end-1]
T = 2*pi/n*sum(1 ./ c.(xx))
```

```
12.825498301618637
```

```
c = x -> 0.2 + sin(x - 1)^2
f = x -> exp(-100*(x-1)^2)
nxv = 2 .^(5:10) .+ 1
maxerr = []
for nx = nxv
    m = (nx-1)÷2
    h = 2*pi/nx
    x = h*(0:nx-1)
    tv = range(0, T; length=8nx+1)
    τ = tv[2]-tv[1]
    u = f.(x)
    uold = f.(x .+ 0.2τ)
    @time begin
    for n = 1:8nx
        unew = real.(uold - 2τ*c.(x).*ifft(ifftshift(im*(-m:m)).*fft(u)))
        uold = u
        u = unew
    end
end
global maxerr
maxerr = [maxerr; norm(u-f.(x), Inf)]
end
```

```

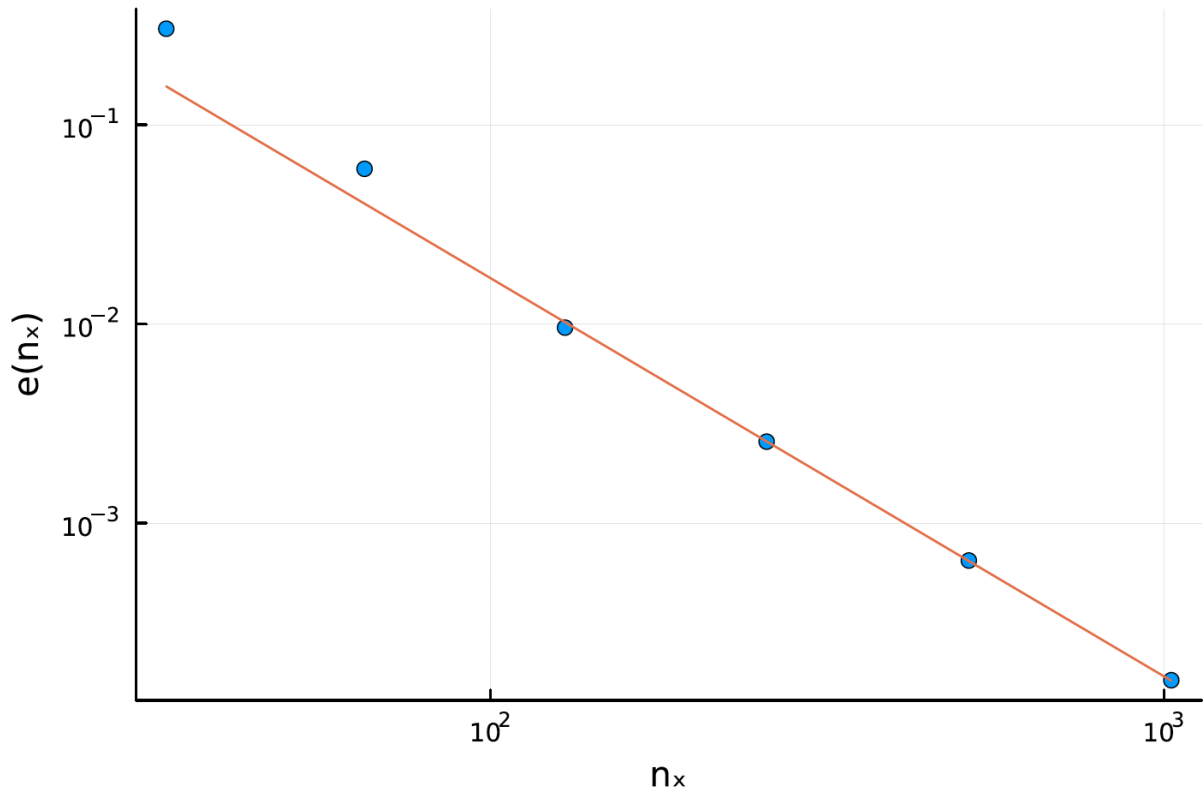
0.096903 seconds (159.31 k allocations: 9.735 MiB, 61.45% compilation time)
0.126623 seconds (38.50 k allocations: 6.919 MiB)
0.151179 seconds (77.41 k allocations: 22.644 MiB)
0.765419 seconds (155.24 k allocations: 80.015 MiB, 6.99% gc time)
1.672286 seconds (310.88 k allocations: 295.999 MiB, 3.96% gc time)
5.037372 seconds (679.58 k allocations: 1.105 GiB, 3.22% gc time)

```

```

scatter(nxv,maxerr;yscale=:log10,xscale=:log10,legend=false,
xlabel="n_x",ylabel="e(n_x)")
plot!(nxv,170 ./nxv.^2)

```



The plot indicates that $e(n_x) = \mathcal{O}(n_x^{-2})$, $n_x \rightarrow \infty$.

The (well-posed) convection-diffusion equation is

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} - b \frac{\partial u}{\partial x}, \quad 0 \leq x \leq 1, \quad 0 \leq t \leq T, \quad T > 0,$$

where $b > 0$ is given (a constant) and the boundary conditions are $u(0, t) = \varphi_0(t)$ and $u(1, t) = \varphi_1(t)$ for $t \in [0, T]$. Let

$$v'_j = \frac{1}{h^2} (v_{j-1} - 2v_j + v_{j+1}) - \frac{b}{2h} (v_{j+1} - v_{j-1}), \quad j = 1, 2, \dots, n_x,$$

where $h = \frac{1}{n_x + 1}$ and $v_j = v_j(t)$ be a semi-discrete method for the convection-diffusion equation.

5. **[5 marks]** Prove that the semi-discrete method is second-order accurate. That is, let $\tilde{v}_j(t) = u(x_j, t)$ and show that (assuming all the relevant partial derivatives are bounded on $(x, t) \in [0, 1] \times [0, T]$),

$$\tilde{v}'_j - \frac{1}{h^2} (\tilde{v}_{j-1} - 2\tilde{v}_j + \tilde{v}_{j+1}) + \frac{b}{2h} (\tilde{v}_{j+1} - \tilde{v}_{j-1}) = \mathcal{O}(h^2), \quad h \rightarrow 0.$$

Solution Using Taylor's theorem (see Chapter 1), there exists a $\xi_+ \in (x_j, x_j + h)$ such that

$$\begin{aligned} \tilde{v}_{j+1} &= u(x_{j+1}, t) \\ &= u(x_j + h, t) \\ &= u(x_j, t) + hu_x(x_j, t) + \frac{h^2}{2}u_{xx}(x_j, t) + \frac{h^3}{6}u_{xxx}(x_j, t) + \frac{h^4}{24}u_{xxxx}(\xi_+, t) \\ &= \tilde{v}_j + hu_x(x_j, t) + \frac{h^2}{2}u_{xx}(x_j, t) + \frac{h^3}{6}u_{xxx}(x_j, t) + \frac{h^4}{24}u_{xxxx}(\xi_+, t). \end{aligned}$$

Similarly, by replacing h with $-h$ in the equations above, there exists a $\xi_- \in (x_j - h, x_j)$ such that

$$\begin{aligned} \tilde{v}_{j-1} &= u(x_{j-1}, t) \\ &= u(x_j - h, t) \\ &= u(x_j, t) - hu_x(x_j, t) + \frac{h^2}{2}u_{xx}(x_j, t) - \frac{h^3}{6}u_{xxx}(x_j, t) + \frac{h^4}{24}u_{xxxx}(\xi_-, t) \\ &= \tilde{v}_j - hu_x(x_j, t) + \frac{h^2}{2}u_{xx}(x_j, t) - \frac{h^3}{6}u_{xxx}(x_j, t) + \frac{h^4}{24}u_{xxxx}(\xi_-, t). \end{aligned}$$

Hence, it follows that

$$\frac{1}{h^2} (\tilde{v}_{j-1} - 2\tilde{v}_j + \tilde{v}_{j+1}) = u_{xx}(x_j, t) + \frac{h^2}{24} (u_{xxxx}(\xi_+, t) + u_{xxxx}(\xi_-, t))$$

and we conclude that (since we assume that u_{xxxx} is bounded)

$$\frac{1}{h^2} (\tilde{v}_{j-1} - 2\tilde{v}_j + \tilde{v}_{j+1}) = u_{xx}(x_j, t) + \mathcal{O}(h^2), \quad h \rightarrow 0. \quad (\star) \quad (1)$$

Also, it follows that

$$\frac{1}{2h} (\tilde{v}_{j+1} - \tilde{v}_{j-1}) = u_x(x_j, t) + \frac{h^2}{6} u_{xxx}(x_j, t) + \frac{h^3}{48} (u_{xxxx}(\xi_+, t) - u_{xxxx}(\xi_-, t))$$

and we conclude that (since we assume that u_{xxx} and u_{xxxx} are bounded)

$$\frac{1}{2h} (\tilde{v}_{j+1} - \tilde{v}_{j-1}) = u_x(x_j, t) + \mathcal{O}(h^2), \quad h \rightarrow 0. \quad (\diamond) \quad (2)$$

Using $(\star)$ and $(\diamond)$ and the advection-diffusion equation, it follows that as $h \rightarrow 0$,

$$\begin{aligned} & \tilde{v}'_j - \frac{1}{h^2} (\tilde{v}_{j-1} - 2\tilde{v}_j + \tilde{v}_{j+1}) + \frac{b}{2h} (\tilde{v}_{j+1} - \tilde{v}_{j-1}) \\ &= \underbrace{u_t(x_j, t) - u_{xx}(x_j, t) + bu_x(x_j, t)}_{=0} + \mathcal{O}(h^2) \\ &= \mathcal{O}(h^2). \end{aligned}$$

6. **[2 marks]** Is the semi-discrete method consistent? Motivate your answer.

Solution Yes, since the method is second-order, the local error tends to zero as $h \rightarrow 0$ and therefore the method is consistent.

7. **[6 marks]** Use the von Neumann method to determine whether the semi-discrete method is stable.

Solution

Substituting

$$v_j(t) = v_j = \lambda(t) e^{ikx_j} = \lambda e^{ikx_j},$$

where $x_j = jh$ and $k \in \mathbb{Z}$, into the semi-discrete method and using the fact that $v_{j+1} = \lambda e^{ikx_{j+1}} = \lambda e^{ik(x_j+h)} = e^{ikh} v_j$, we find that

$$\begin{aligned} v'_j &= \frac{1}{h^2} (e^{-ikh} - 2 + e^{ikh}) v_j - \frac{b}{2h} (e^{ikh} - e^{-ikh}) v_j \\ &= \frac{1}{h^2} (2 \cos(kh) - 2) v_j - \frac{b}{2h} (2i \sin(kh)) v_j \\ &= -\frac{2}{h^2} (1 - \cos(kh)) v_j - \frac{bi}{h} \sin(kh) v_j \\ &= -\left[\frac{4}{h^2} \sin^2(kh/2) + \frac{bi}{h} \sin(kh) \right] v_j \\ \implies \lambda' &= -\left[\frac{4}{h^2} \sin^2(kh/2) + \frac{bi}{h} \sin(kh) \right] \lambda \\ \implies \lambda(t) &= \exp\left(-\left[\frac{4}{h^2} \sin^2(kh/2) + \frac{bi}{h} \sin(kh) \right] t\right) \lambda(0). \end{aligned}$$

Since, as $h \rightarrow 0$,

$$\frac{4}{h^2} \sin^2(kh/2) = k^2 + \mathcal{O}(h^2), \quad \frac{1}{h} \sin(kh) = k + \mathcal{O}(h^2),$$

we have that for $t \in [0, T]$,

$$\begin{aligned} \lim_{h \rightarrow 0} |\lambda(t)| &= \lim_{h \rightarrow 0} \left| \exp \left(- \left[\frac{4}{h^2} \sin^2(kh/2) + \frac{bi}{h} \sin(kh) \right] t \right) \lambda(0) \right| \\ &= \lim_{h \rightarrow 0} \exp \left(- \frac{4}{h^2} \sin^2(kh/2) t \right) |\lambda(0)| \\ &= e^{-k^2 t} |\lambda(0)| \\ &\leq |\lambda(0)| \\ &< \infty \end{aligned}$$

hence the method is stable.

8. **[2 marks]** Is the semi-discrete method convergent?

Solution Yes, since the advection-diffusion equation is well-posed and the semi-discrete method is consistent and stable, it is convergent by the Lax equivalence theorem.

9. **[5 marks]** Show that the semi-discrete method can be expressed as the following system of ordinary differential equations (ODEs):

$$\mathbf{v}' = A\mathbf{v} + \mathbf{h}$$

with $\mathbf{v}, \mathbf{h} \in \mathbb{R}^{n_x}$, $A \in \mathbb{R}^{n_x \times n_x}$ where

$$A = \frac{1}{h^2} \begin{bmatrix} -2 & 1 - \frac{bh}{2} & & & \\ 1 + \frac{bh}{2} & -2 & 1 - \frac{bh}{2} & & \\ & \ddots & \ddots & \ddots & \\ & & 1 + \frac{bh}{2} & -2 & 1 - \frac{bh}{2} \\ & & & 1 + \frac{bh}{2} & -2 \end{bmatrix}, \quad \mathbf{h} = \frac{1}{h^2} \begin{bmatrix} \left(1 + \frac{bh}{2}\right) \varphi_0(t) \\ 0 \\ \vdots \\ 0 \\ \left(1 - \frac{bh}{2}\right) \varphi_1(t) \end{bmatrix}$$

Solution Rewriting the semi-discrete method

$$\begin{aligned} v_j' &= \frac{1}{h^2} (v_{j-1} - 2v_j + v_{j+1}) - \frac{b}{2h} (v_{j+1} - v_{j-1}) \\ &= \frac{1}{h^2} \left[\left(1 + \frac{bh}{2}\right) v_{j-1} - 2v_j + \left(1 - \frac{bh}{2}\right) v_{j+1} \right], \quad j = 1, \dots, n_x, \end{aligned}$$

it follows that

$$\mathbf{v}' = A\mathbf{v} + \mathbf{h}$$

where

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_{n_x} \end{pmatrix}, \quad A = \frac{1}{h^2} \begin{bmatrix} -2 & 1 - \frac{bh}{2} & & & \\ 1 + \frac{bh}{2} & -2 & 1 - \frac{bh}{2} & & \\ & \ddots & \ddots & \ddots & \\ & & 1 + \frac{bh}{2} & -2 & 1 - \frac{bh}{2} \\ & & & 1 + \frac{bh}{2} & -2 \end{bmatrix}, \quad \mathbf{h} = \frac{1}{h^2} \begin{pmatrix} \left(1 + \frac{bh}{2}\right) \varphi_0(t) \\ 0 \\ \vdots \\ 0 \\ \left(1 - \frac{bh}{2}\right) \varphi_1(t) \end{pmatrix}$$

10. **[10 marks]** Let $n_x = 300$, $T = 0.003$, $b = 100$, $\varphi_0(t) = 0 = \varphi_1(t)$ and $u(x, 0) = e^{-300(x-0.3)^2}$, then solve the ODE system in question 5 with an error tolerance of 10^{-4} using any ODE solver that's available in your programming language of choice. Plot the solution at $t = 0$ and $t = T$ on the same set of axes.

Solution

```
using LinearAlgebra, Plots, OrdinaryDiffEq

nx = 300
x = range(0,1,nx+2)
f = x -> exp(-300*(x-0.3).^2)
h = 1/(nx+1)
T = 0.003
b = 100
A = Tridiagonal(fill(1+b*h/2,nx-1),fill(-2.0,nx),fill(1-b*h/2,nx-1))/h^2
F = (v,p,t) -> A*v
x = range(0,1,nx + 2)
prob = ODEProblem(F, f.(x[2:end-1]), (0.0, T))
#soln = solve(prob, RK4(), abstol=1e-4)
#soln = solve(prob, Rodas4(), abstol=1e-4);
soln = solve(prob, Rodas4P(), abstol=1e-4)
@show nt = length(soln.t)
plot(x[2:nx+1],soln.u[nt],label = "t = 0.003")
plot!(x[2:nx+1],soln.u[1],label = "t = 0")

nt = length(soln.t) = 14
```

